TRYHACKME WIRESHARK TRAFFIC ANALYSIS

SOC Level 1 • Network Forensics Investigation Report

Assessment type	Educational SOC lab / packet-capture analysis
Analysis host	192.168.31.172 (Kali Linux, based on supplied screenshot)
Target host	192.168.31.104 (Metasploitable, based on supplied screenshot)
Tools	Wireshark, Nmap, Linux command line
Evidence scope	Supplied screenshots only; no live retesting performed

Executive summary. The supplied packet evidence shows broad service exposure on the target, a successful FTP login with credentials visible in cleartext, HTTP access to /dav/, SMB IPC$ / SRVSVC enumeration, ICMP reachability, and TCP SYN/RST activity. These are meaningful defensive findings, but the screenshots do not prove that a remote attacker exploited a specific service or obtained operating-system control.

Overall assessment: High-risk exposure is indicated by cleartext FTP authentication and multiple legacy services on an intentionally vulnerable host. Severity ratings below are provisional and scoped to the evidence shown.

1. Scope and methodology

This report follows the Kioptrix-style finding format: a separate table for each finding, with port/service, severity, vulnerability name, CVE status, description, proof of concept, and remediation. Only the most useful screenshots are included. Findings are based on the uploaded images; packet capture files were not provided for independent re-analysis.

The screenshots identify 192.168.31.172 as the Kali-side host and 192.168.31.104 as the Metasploitable-side target. This report treats 192.168.31.104 as the target because it is the host scanned and receiving the shown HTTP/SMB/ICMP traffic.

2. Executive findings

#	Port / activity	Finding	Severity
1	21/TCP FTP	Successful FTP login; username and password visible in plaintext	High
2	Multiple TCP services	Large exposed service surface including Telnet, SMB, bind shell and IRC	High (provisional)
3	445/TCP SMB	IPC$ connection and SRVSVC share enumeration observed	Medium
4	80/TCP HTTP	GET request to /dav/ observed; endpoint exposure requires configuration review	Informational / Medium
5	TCP/80 scan-like traffic	Repeated SYN probes and RST responses in capture	Informational

3. Findings — Kioptrix-style PoC tables

Finding 1 — Cleartext FTP credentials

FIELD	DETAILS
Port / service	21/tcp — FTP (banner shown as vsFTPd 2.3.4 in the supplied Nmap screenshot; verify against the raw scan before relying on version-specific claims)
Severity	High
Finding	Credentials transmitted without encryption
CVE ID	N/A — not established by supplied evidence
Description	Wireshark Follow TCP Stream shows `USER msfadmin`, `PASS msfadmin`, followed by `230 Login successful.` The username and password are visible in the application-layer FTP conversation. This is direct evidence of cleartext credential exposure in the captured session. The evidence does not prove the same credentials are valid outside this lab.
Proof of concept / validation	In the supplied capture, use `ftp` as the display filter and select the relevant FTP conversation. Follow the TCP stream and confirm the USER/PASS commands and 230 response. Do not reproduce these credentials against systems outside the authorized lab.
Remediation	Disable FTP where unnecessary. Prefer SFTP or FTPS with correctly validated TLS. Rotate credentials that were transmitted in cleartext, enforce unique strong passwords, restrict FTP access by network, and monitor successful/failed authentication.

Finding 2 — Excessive exposed services / attack surface

FIELD	DETAILS
Port / service	21, 22, 23, 25, 53, 80, 111, 139, 445, 512–514, 1099, 1524, 2049, 2121, 3632, 5900, 6000, 6667, 6697, 8787, 39216, 46603, 52540, 59429/tcp (as shown by Nmap)
Severity	High (provisional; lab context)
Finding	Numerous network services exposed on one host
CVE ID	N/A — not established by supplied evidence

FIELD	DETAILS
Description	Nmap service detection reports many open services, including FTP, SSH, Telnet, SMTP, DNS, HTTP, RPC, SMB, a bind shell on 1524/tcp, NFS, VNC and IRC. The bind shell and legacy cleartext services deserve immediate attention. Service banners alone do not confirm that every listed service is exploitable.
Proof of concept / validation	In the authorized lab only, reproduce service discovery with `nmap -sS -sV -O 192.168.31.104`. Save the output and validate open ports against the screenshot. Avoid exploit attempts unless separately authorized.
Remediation	Disable unused services, isolate the lab host, restrict ingress with host/network firewalls, remove bind shells and legacy daemons, patch supported services, and require explicit service ownership. Re-scan after remediation.

Finding 3 — SMB IPC$ and service enumeration

FIELD	DETAILS
Port / service	139/tcp and 445/tcp — NetBIOS/SMB
Severity	Medium
Finding	SMB share/service enumeration visible in traffic
CVE ID	N/A — not established by supplied evidence
Description	The screenshot shows an SMB tree connection to `\\192.168.31.104\IPC$`, followed by an SRVSVC `NetShareEnumAll` request. This is consistent with share enumeration. The supplied evidence does not show a successful file read, write, or remote code execution.
Proof of concept / validation	Apply the Wireshark display filter `smb || ntlmssp`. Inspect Tree Connect requests and SRVSVC traffic. Confirm the target path `\\192.168.31.104\IPC$` and the share-enumeration request in the packet details.
Remediation	Disable SMBv1 and unused SMB services, patch the server, restrict TCP/139 and TCP/445 to required hosts, disable guest/anonymous access, enforce least privilege on shares, and monitor unusual enumeration.

Finding 4 — HTTP WebDAV path request

FIELD	DETAILS
Port / service	80/tcp — HTTP (Apache httpd 2.2.8, as reported by Nmap)
Severity	Informational to Medium; configuration-dependent
Finding	Request to `/dav/` observed
CVE ID	N/A — not established by supplied evidence
Description	The HTTP capture shows a GET request to `/dav/` from 192.168.31.172 to 192.168.31.104. This confirms that the path was requested, not that WebDAV was enabled, writable, or vulnerable. No specific CVE is assigned from this evidence alone.
Proof of concept / validation	Use `http.request.method == "GET"` or `http.request.uri contains "/dav/"` in Wireshark. Review the corresponding HTTP response and server configuration before classifying the endpoint as vulnerable.
Remediation	Disable WebDAV if not required. If required, restrict methods and access, require authentication, prevent write access where unnecessary, patch Apache, and review access/error logs.

Finding 5 — TCP SYN probing and reset responses

FIELD	DETAILS
Port / service	TCP/80 — repeated SYN packets and RST responses

FIELD	DETAILS
Severity	Informational; activity classification requires context
Finding	Connection-probing / scan-like traffic
CVE ID	N/A — not established by supplied evidence
Description	The supplied screenshots show repeated TCP SYN packets aimed at port 80 on 192.168.31.104 and TCP RST packets later in the trace. This is consistent with probing or connection attempts. A SYN burst or RST alone does not establish malicious intent, a successful connection, or the source of the activity.
Proof of concept / validation	Use `tcp.flags.syn == 1 && tcp.flags.ack == 0` to inspect initial SYNs and `tcp.flags.reset == 1` to inspect resets. Compare source/destination addresses, ports, timing, and subsequent handshake packets before attributing the activity.
Remediation	Use network monitoring and rate limits where appropriate, restrict exposed ports, alert on high-volume connection attempts, and correlate with firewall/IDS logs. Avoid blocking addresses solely on the basis of this screenshot without validating attribution.

4. Proof-of-concept / validation commands

Use these only inside the authorized TryHackMe/local lab. The commands validate exposure and evidence; they do not exploit the target.

Service enumeration

```
nmap -sS -sV -O 192.168.31.104
```

FTP session validation (use only the lab credentials shown in your own lab)

```
ftp 192.168.31.104
# In Wireshark, use display filter: ftp
# Follow the relevant TCP stream and verify whether USER/PASS are visible
```

HTTP request review

```
http.request.method == "GET"
http.request.uri contains "/dav/"
```

SMB review

```
smb || ntlmssp
smb contains "IPC$"
```

TCP SYN and reset review

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
tcp.flags.reset == 1
```

ICMP reachability review

```
icmp
```

5. Attack reconstruction and conclusion

The screenshots support the following evidence-based sequence: (1) the lab analyst identified the target and confirmed ICMP reachability; (2) Nmap enumerated a broad set of open services; (3) the FTP conversation exposed a successful login with credentials in cleartext; (4) SMB traffic included IPC$ and SRVSVC share enumeration; (5) an HTTP request targeted `/dav/`; and (6) the capture included repeated TCP SYN probes and reset packets.

Conclusion: The clearest confirmed security issue is cleartext FTP authentication, which allows anyone able to capture the relevant traffic to read the username and password. The target also exposes a large legacy service surface, including Telnet, SMB, a bind shell, and IRC, increasing opportunities for attack if reachable by untrusted hosts. SMB enumeration and HTTP `/dav/` access are confirmed as observed activity, but the screenshots do not establish successful exploitation. Treat the SYN/RST traffic as scan-like until the complete PCAP, timing, and host context confirm intent.

Priority mitigations: (1) replace FTP with SFTP/FTPS and rotate any exposed credentials; (2) remove the bind shell and disable unnecessary/legacy services; (3) isolate the target and restrict inbound ports; (4) harden SMB and review share permissions; (5) verify whether WebDAV is enabled and disable it if unnecessary; and (6) correlate packet evidence with host and firewall logs before making attribution claims.

6. Limitations

This is a screenshot-based analysis. The original PCAP/PCAPNG files were not supplied, so packet counts, full TCP handshakes, FTP file transfers, SMB authorization outcomes, HTTP response status, and exact scan origin could not be independently verified. Version-specific exploitability is not claimed from banners alone.

7. Online references

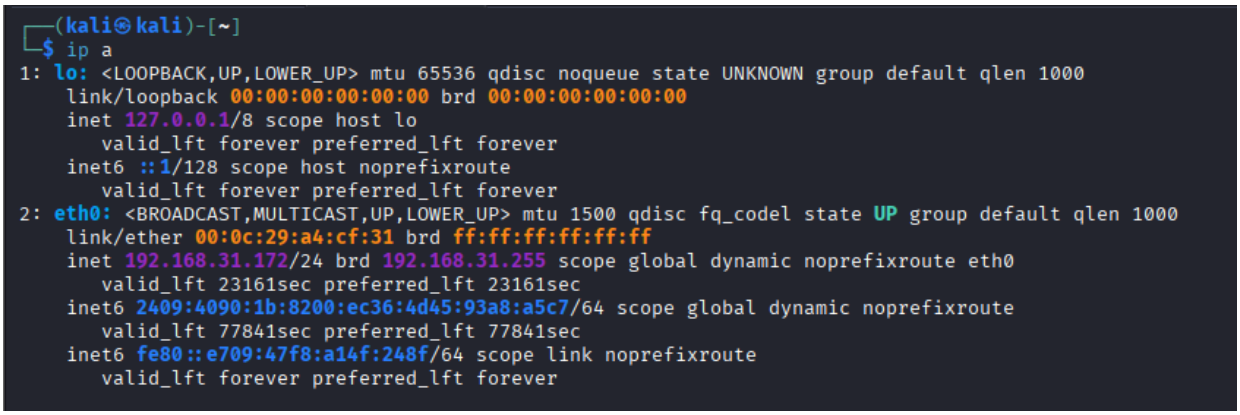
TryHackMe — Wireshark: Traffic Analysis: https://tryhackme.com/room/wiresharktrafficanalysis

Jalblas — TryHackMe Wireshark Traffic Analysis Walkthrough: https://www.jalblas.com/blog/thm-wireshark-traffic-analysis-walkthrough/

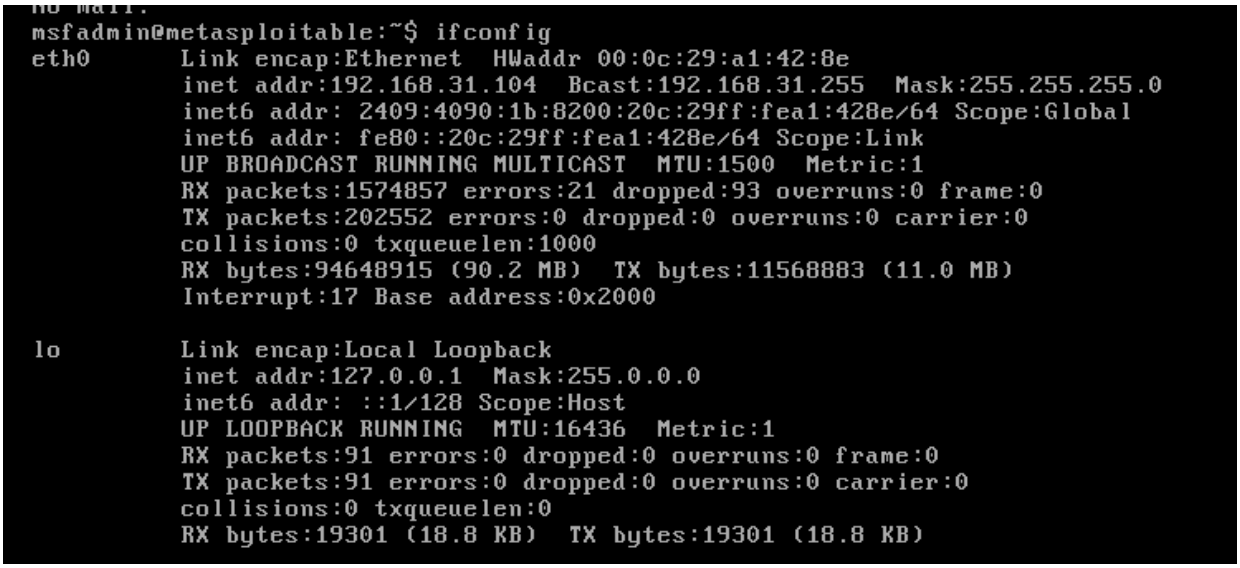
GitHub — Wireshark Traffic Analysis write-up: https://github.com/veyronshade/tryhackme-writeups/blob/main/cybersecurity-101/Wireshark-Room/Wireshark%20Traffic%20Analysis.md

These references provide room context and general walkthrough structure; the findings in this report are grounded in the supplied screenshots.

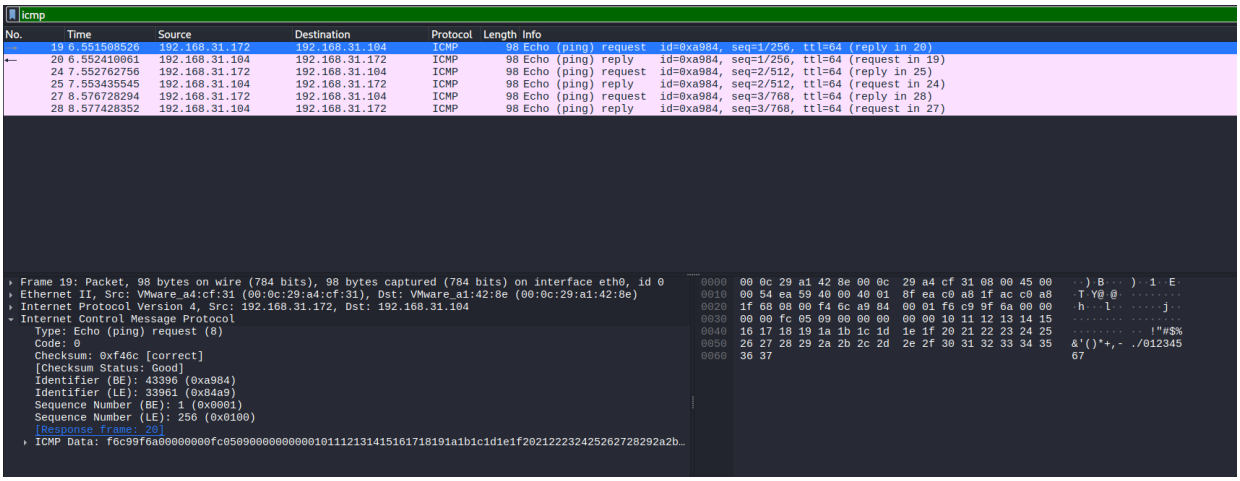
Appendix A — Selected screenshot evidence



Evidence 01 — Kali interface configuration. The attacker/lab analysis host is shown as 192.168.31.172/24.



Evidence 02 — Metasploitable interface configuration. The target is shown as 192.168.31.104/24.



Evidence 03 — ICMP echo requests and replies between 192.168.31.172 and 192.168.31.104 confirm reachability during the capture.

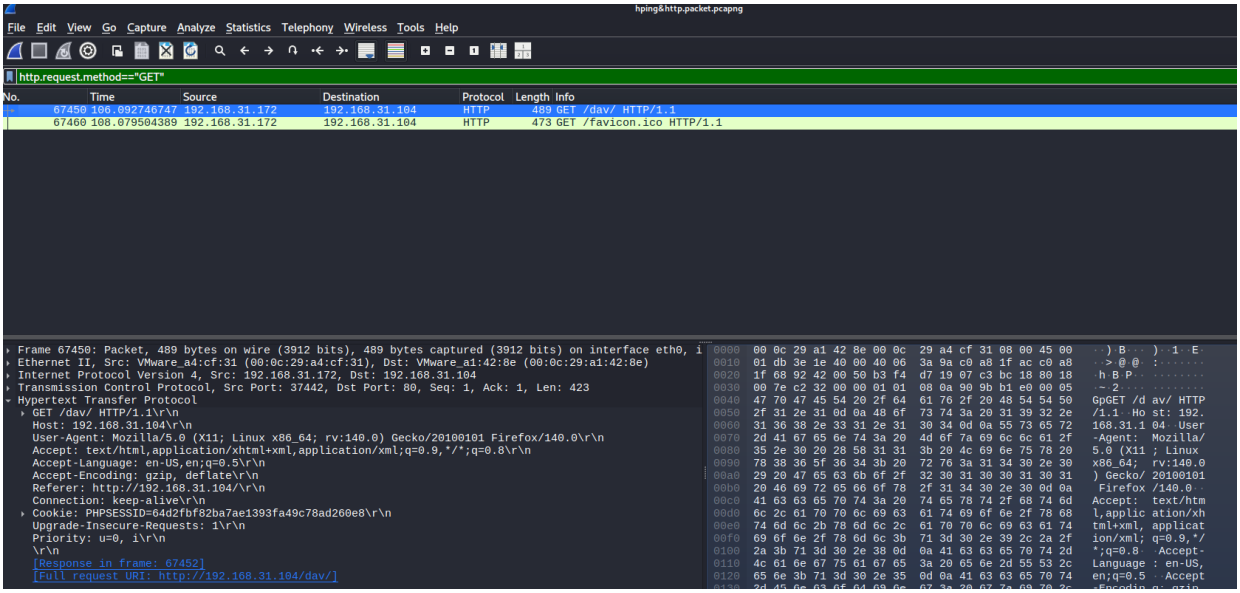
```
┌──(kali㉿kali)-[~/Wireshark]
└─$ nmap -sS -sV -O -p- 192.168.31.104
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-08 14:36 +0500
Nmap scan report for 192.168.31.104
Host is up (0.0014s latency).
Not shown: 65509 closed tcp ports (reset)
PORT      STATE SERVICE     VERSION
21/tcp    open  ftp         vsftpd 2.3.4
22/tcp    open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet      Linux telnetd
25/tcp    open  smtp        Postfix smtpd
53/tcp    open  domain      ISC BIND 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind     2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec?
513/tcp   open  login       OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi    GNU Classpath grmiregistry
1524/tcp  open  bindshell   Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ftp         ProFTPD 1.3.1
3632/tcp  open  distccd     distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5900/tcp  open  vnc         VNC (protocol 3.3)
6000/tcp  open  X11         (access denied)
6667/tcp  open  irc         UnrealIRCd
6697/tcp  open  irc         UnrealIRCd
8787/tcp  open  drb         Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drb)
39216/tcp open  java-rmi    GNU Classpath grmiregistry
46603/tcp open  mountd      1-3 (RPC #100005)
52540/tcp open  status      1 (RPC #100024)
59429/tcp open  nlockmgr    1-4 (RPC #100021)
MAC Address: 00:0C:29:A1:42:8E (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
```

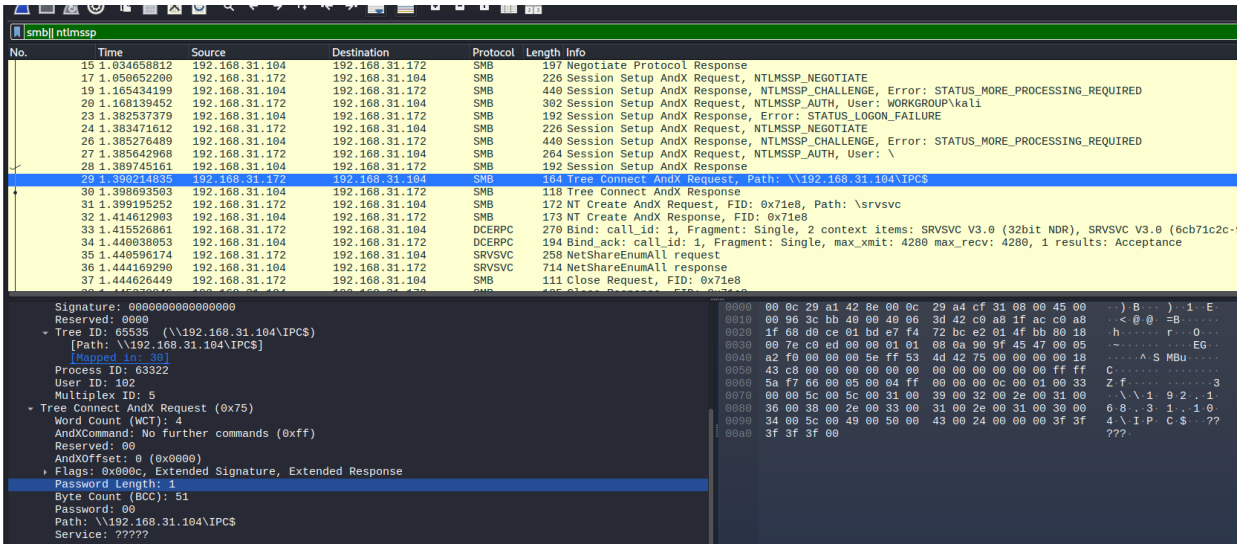
Evidence 04 — Nmap service detection against 192.168.31.104 lists multiple open services, including FTP, SSH, Telnet, HTTP, SMB, bind shell, and IRC.



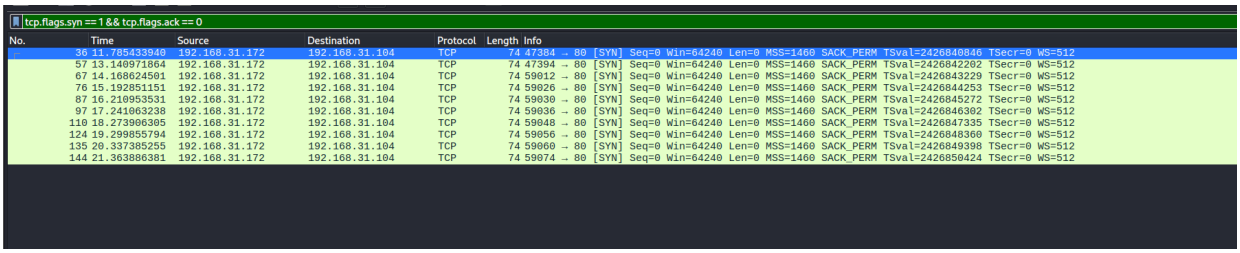
Evidence 05 — FTP Follow TCP Stream shows USER msfadmin, PASS msfadmin, and a 230 Login successful response. FTP credentials are transmitted in cleartext.



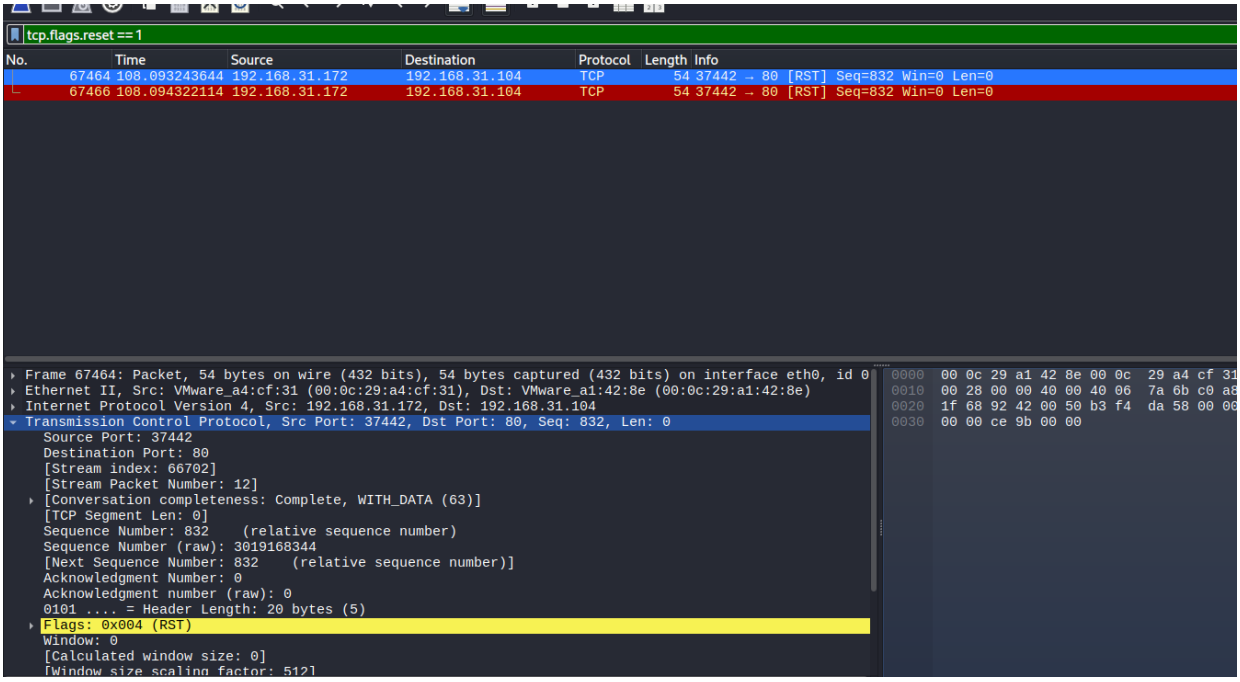
Evidence 06 — HTTP display filter shows GET /dav/ and GET /favicon.ico requests from 192.168.31.172 to 192.168.31.104.



Evidence 07 — SMB traffic shows an IPC$ tree connection and SRVSVC NetShareEnumAll request, consistent with share enumeration.



Evidence 08 — TCP SYN-only packets target TCP/80 on 192.168.31.104. This is consistent with connection probing/scanning; source attribution and intent require full capture context.



Evidence 09 — TCP RST packets are visible after HTTP connection attempts. Resets indicate connection termination/rejection, not by themselves proof of an attack.